

Exercises 1.8

1. Let $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(x) = Ax$. Find the images under T of $u = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$ and $v = \begin{bmatrix} a \\ b \end{bmatrix}$.

Solution: $T(u) = Au = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \cdot 1 + 0 \cdot (-3) \\ 0 \cdot 1 + 2 \cdot (-3) \end{bmatrix} = \begin{bmatrix} 2 \\ -6 \end{bmatrix}$
(By Row-Vector Rule)

$T(v) = Av = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2 \cdot a + 0 \cdot b \\ 0 \cdot a + 2 \cdot b \end{bmatrix}$ (Row-vector Rule) $= \begin{bmatrix} 2a \\ 2b \end{bmatrix}$

3. In Exercises 3-6, with T defined by $T(x) = Ax$, find a vector x whose image under T is b , and determine whether x is unique.

$A = \begin{bmatrix} 1 & 0 & -2 \\ -2 & 1 & 6 \\ 3 & -2 & -5 \end{bmatrix}$, $b = \begin{bmatrix} -1 \\ 7 \\ -3 \end{bmatrix}$

Solution: We need to find the vector x s.t. $T(x) = b$. $\therefore Ax = b$
By Theorem 3, the matrix equation $Ax = b$ has the same solution as the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{aligned} &\begin{bmatrix} 1 & 0 & -2 & -1 \\ -2 & 1 & 6 & 7 \\ 3 & -2 & -5 & -3 \end{bmatrix} \xrightarrow[\text{Row 3} = \text{Row 3} + 2 \times \text{Row 2}]{\text{Row 2} = \text{Row 2} + 2 \times \text{Row 1}} \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 5 \\ 3 & -2 & -5 & -3 \end{bmatrix} \xrightarrow[\text{Row 3} = \text{Row 3} - 3 \times \text{Row 1}]{\text{Row 3} = \text{Row 3} - 3 \times \text{Row 1}} \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & -2 & 1 & 0 \end{bmatrix} \\ &\xrightarrow[\text{Row 2} = \text{Row 2} - 2 \times \text{Row 3}]{\text{Row 3} = \text{Row 3} \times 1/5} \begin{bmatrix} 1 & 0 & -2 & -1 \\ 0 & 1 & 2 & 5 \\ 0 & 0 & 1 & 2 \end{bmatrix} \xrightarrow[\text{Row 1} = \text{Row 1} + 2 \times \text{Row 3}]{\text{Row 2} = \text{Row 2} - 2 \times \text{Row 3}} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix} \end{aligned}$$

Verification of whether the final matrix is in rref

1. All the rows of the matrix are non-zero rows. $\therefore$ The condition that the non-zero rows must be above the zero rows is vacuously True.
2. Row 1 Leading entry = 1 in Column 1
Row 2 Leading entry = 1 in Column 2
Row 3 Leading entry = 1 in Column 3.